Surname : Solution Guide Initials :

Student number:

STICK BARCODE STICKER HERE

We penalise for:

1. Missing units
2. Inconsistent logic and/or notation
3. No sketches especially FBD's
4. Missing coordinate systems

✓ = 1 mark ✓ = ½ mark.

FSK 116

Semester Test 1

Department of Physics, University of Pretoria

Date: 2024-04-10 **Time: 3.0 hours** **Total: 80**

Examiner: RQ Odendaal **Moderator: Prof JM Nel**

All calculation steps are not shown.

✂ ········ ✂ ········ ✂ ········ ✂ ········ ✂

Test register: Tear carefully on the line above and hand to the invigilator.

FSK 116: Semester Test 1 **2024-04-10**

Surname					Initials			
Student number								
Venue								
Signature								

STICK BARCODE STICKER HERE

Please read the following carefully.

1. Answer all the questions in the space provided on the question paper.
2. No pencil work will be marked. Do not use "tippex".
3. If you need more space, please use the back of the pages, and clearly indicate where the question continues.
4. Where applicable you need to draw the necessary diagrams (e.g. free-body diagrams).
5. Your answer needs to be written in a systematic and logical fashion. You must show that you understand and can use the relevant physics. **Only substitute numerical values in the final step of calculations! A correct answer does not imply that you will receive full marks.**
6. **Units must be used when numbers are substituted else marks will be subtracted.**
7. The use of the cheat sheet is only allowed subject to the rules stated on it.
8. All exam regulations of the University of Pretoria apply, and any contravention of it will be treated in a serious manner.

Constants and formulas

$\varepsilon_0 = 8.854 \times 10^{-12}$ C^2/(N·m^2); $G = 6.674 \times 10^{-11}$ N·m^2/kg^2; $e = 1.602 \times 10^{-19}$ C; $k = 1.381 \times 10^{-23}$ J/K; $m_e = 9.109 \times 10^{-31}$ kg; $m_p = 1.673 \times 10^{-27}$ kg; $R = 8.314$ J/(mol·K); $N_A = 6.02 \times 10^{23}$ /mol; 1 atm = 101.3 kPa; 1 u = $1.6605677 \times 10^{-27}$ kg; $c = 299\ 792.5$ km/s; $V_{\text{sphere}} = \frac{4}{3}\pi r^3$; $F_{\text{gravity}} = \frac{Gm_1m_2}{r^2}$; 1 litre = 1000 cm^3; $M_{\text{Earth}} = 5.97 \times 10^{24}$ kg, ρ(water) = 1 g/cm^3; $r_{\text{Earth}} = 6.37 \times 10^6$ m; $h = 6.626 \times 10^{-34}$ J·s

---

1

Question 1 [11]

In fundamental Physics, there is a time defined as the Planck time, t_P. One Planck time is the time it would take a photon travelling at the speed of light to cross a distance equal to one Planck length (another fundamental quantity).

The Planck time is defined as:

$$t_P = \sqrt{\frac{hG}{2\pi c^5}},$$

where h is the Planck constant, G is Newton's gravitational constant, and c is the speed of light in a vacuum. (Page 2 lists the values of each of these constants.)

(a) Show the dimensions of each constant in the equation above, and show that the combination has indeed a dimension of time. (5)

5

$[\text{Force}] = [MLT^{-2}]$

$[\text{Energy}] = [\text{Force} \times L]$

$[G] = [\text{Force} \times L^2 \times M^{-2}]$ ✓   ✓ $[2\pi]$ is dimensionless

$[h] = [\text{Energy} \times T]$ ✓

$[c] = [\frac{L}{T}] = [L \times T^{-1}]$ ✓   ✓ Simplification must be shown.

$[\frac{hG}{c^5}] = [T^2] \Rightarrow [\sqrt{\frac{hG}{2\pi c^5}}] = T$

(b) Calculate the value of one Planck time in femtoseconds. You need to show each conversion factor that you use in the calculation. If you do not show a multiplicative conversion factor, you will be penalised. (6)

$$\sqrt{\frac{hG}{2\pi c^5}} = \left[\frac{(6.626\times10^{-34}\ J\cdot s)(6.674\times10^{-11}\ Nm^2\cdot kg^{-2})}{2\pi\,(299\,792.5\ km\cdot s^{-1})^5}\times\frac{1N\cdot m}{1J}\right.$$

$$\left.\times\left(\frac{1kg\cdot m\cdot s^{-2}}{1N}\right)^2\times\left(\frac{1km}{10^3\ m}\right)^5\right]^{\frac{1}{2}}$$

✓ ✓

$= \dots\dots$ ✓ simplification.

$= \dots\dots$ ✓

$= 5.391\times10^{-44}\ s\times\frac{1fs}{10^{-15}s}$

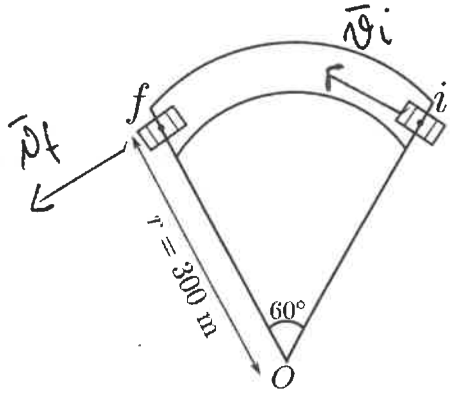
$= 5.391\times10^{-29}\ fs$ ✓

must be shown as factors!

b

2 **Question 2** [8]

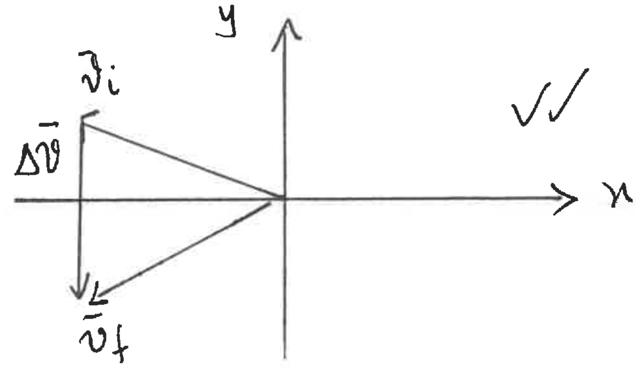
A car moves anticlockwise around a curve of radius 300 m at a **constant** speed of 60 m/s as shown in the figure.



(a) Calculate the resultant change in the velocity when the car goes around an arc of 60°. Hint: Sketch the velocity vectors at the initial and final positions. (4)

4

$\Delta\bar{v} = \bar{v}_f - \bar{v}_i$



$|\bar{v}_i| = |\bar{v}_f|$

$\Delta\vec{v} = 60\,m/s$ ✓ in $-\hat{j}$ ✓ direction

(b) Compare the magnitude of the instantaneous acceleration of the car to the magnitude of the average acceleration over the $60°$ arc. (4)

4

Instantaneous acc. $a_c = \frac{v^2}{r} = \dots = 12\,m/s^2$ ✓

= Centripetal.

Time between i and f : $\Delta t = \frac{\Delta S}{v} = \frac{r\Delta\theta}{v} = \frac{300m\left(\frac{\pi}{3}\right)}{60m/s} = \dots = 5.24s$ ✓ ✓

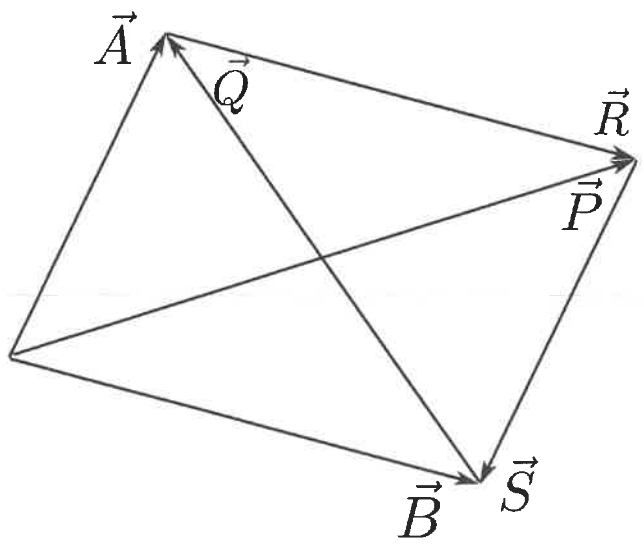
$|\bar{a}_{avg}| = \frac{|\Delta\vec{v}|}{\Delta t} = \dots = 11.46\,m/s$ ✓

$a_c > |\bar{a}_{avg}|$

5

(3) Question 3 [5]

Refer to the figure below. Express the vectors $\vec{P}$, $\vec{R}$, $\vec{S}$, and $\vec{Q}$ in terms of vectors $\vec{A}$ and $\vec{B}$.



$\bar{P} = \bar{A} + \bar{B}$

$\bar{R} = \bar{B}$

$\bar{Q} = \bar{A} - \bar{B}$

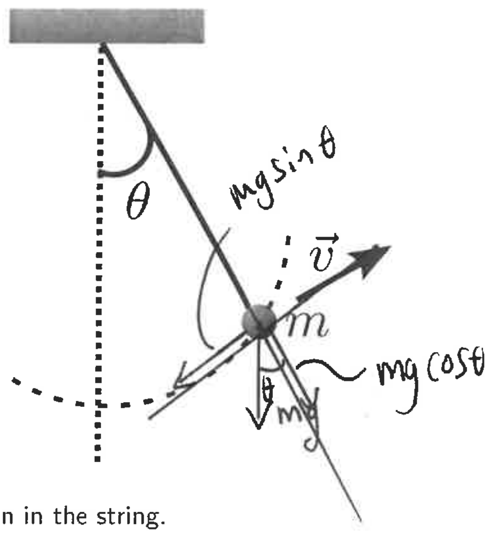
$\bar{S} = -\bar{A}$

Question 4

4

[7]

One end of a cord is fixed and a small ball of mass, $m = 0.500$ kg, is attached to the other end, where it swings in a section of a **vertical** circle of radius 2.00 m as shown in the figure. When $\theta = 20°$, the speed of the object is $v = 8.00$ m/s, with the direction as indicated.



(a) At this instant, find the tension in the string. (2)

2

Radial direction : $\Sigma F_r = ma_r = ma_c$

$T - mg\cos\theta = \frac{mv^2}{r}$ ✓

⋮

$T = 20.6\text{N}$ ✓

(b) At this instant, find the tangential and radial components of the acceleration, as well as the total acceleration of the ball. (5)

5

$a_r = \frac{v^2}{r} = \ldots = 32.0\,m/s^2$ inwards ✓

Tangential direction: $\sum F_t = m a_t$

$mg\sin\theta = m a_t$ ✓

$a_t = 3.35\,m/s^2$ down the tangent line ✓

Total acc. $a = \sqrt{a_t^2 + a_c^2} = \ldots = 32.2\,m/s^2$

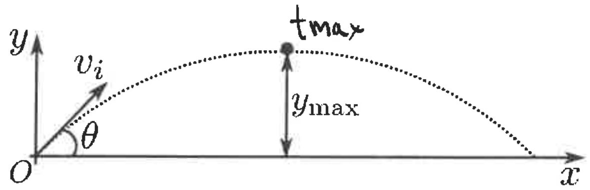
at an angle $\arctan\left(\frac{3.35\,m/s^2}{32\,m/s^2}\right) = 5.98^\circ \simeq 6^\circ$ below the cord.

5 Question 5 [7]

7

The figure shows a projectile being launched from the origin O of the coordinate system. The projectile is launched at an angle of θ with an initial speed v_i. If the position of the projectile is given by $\vec{r} = \vec{r}_i + \vec{v}_i t + \frac{1}{2}\vec{a}t^2$ at any time t, and the velocity of it is given as $\vec{v} = \vec{v}_i + \vec{a}t$ for any time t, show that the maximum height, $y_{\max}$, is given by

$$\frac{v_i^2 \sin^2\theta}{2g}.$$



You need to start with the given equations; you need the y-eq's:

$y = y_i + v_{iy}t + \frac{1}{2}a_y t^2$ and $v_y = v_{iy} + a_y t$

which for the coordinate system changes to

$= 0 + (v_i \sin\theta)t - \frac{1}{2}gt^2$ —① $v_y = v_{iy}\sin\theta - gt$ —②

$v_y = 0$ at t_{max} ⇒ turning point

Find t_{max} from ② and substitute into ① and set $y = y_{max}$.

Simplify to get the asked for result.

We mark proof very strictly: inconsistent notation, unclear logic etc. are penalised.

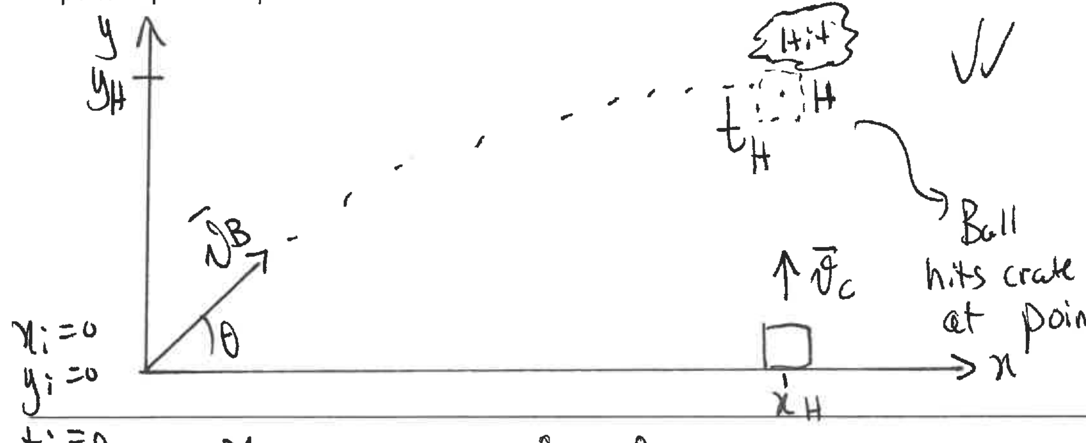
⑥ Question 6 [13]

13

Students at a contest use a compressed-air cannon to shoot a ball at a crate being hoisted upwards at $v_C = 10$ m/s by a crane. The cannon is tilted at a launch angle $\theta = 30°$ above the horizontal and is a distance $d = 100$ m horizontally away from the initial position of the crate. The cannon fires by remote control the instant the crate leaves the ground. The launch speed of the ball, v_B, can be controlled by changing the air pressure in the cannon.

What launch speed, v_B, should the students use for the ball to hit the crate?

Sketch the problem and make sure to indicate your choice of coordinate system. Make sure to label each important point in space.



✓✓

Ball hits crate at point H at time t_H

Coordinate system choice MUST be shown

$x_H = 100$ m $\qquad \theta = 30°$

We find t_H from: $x_H = v_{B,x} t_H$ ✓

$$\Rightarrow t_H = \frac{x_H}{v_B \cos\theta} \quad — ①$$

Crate is at height y_H at time t_H

$$y_{crate,H} = v_C t_H \quad -(2)$$

The ball is at height y_H at time t_H

$$y_{B,H} = y_i + v_{B,y} t_H - \tfrac{1}{2} g t_H^2$$

$$y_{B,H} = 0 + (v_B \sin\theta) t_H - \tfrac{1}{2} g t_H^2 \quad -(3)$$

But $y_{B,H} = y_{crate,H}$ $\Rightarrow$ Set (2) and (3) equal to one another.

$$v_C t_H = (v_B \sin\theta) t_H - \tfrac{1}{2} g t_H.$$

t_H is unknown, substitute from (1), and solve for v_B. v_B is now in a quadratic eq. with solutions

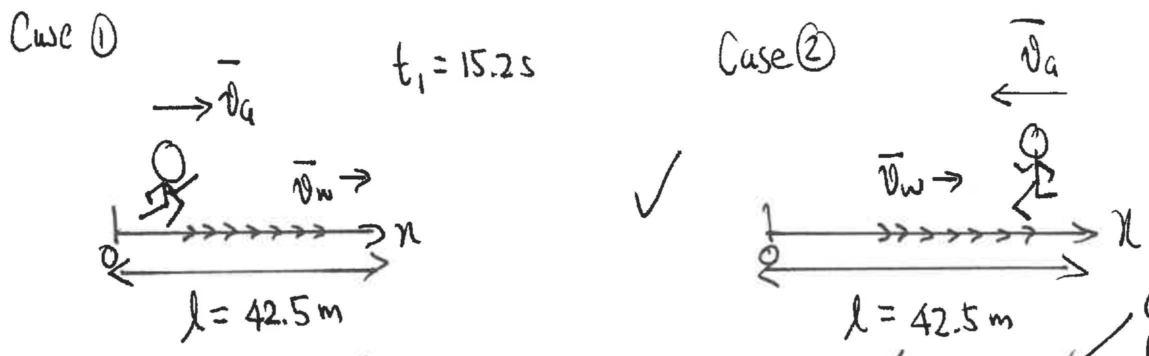
$$v_B = -25.1 \text{ m/s} \quad \text{and} \quad v_B = +45.1 \text{ m/s}$$

non-physical. speed required.

7

7 Question 7 [7]

During a long airport layover, a physicist father and his young daughter try a game that involves a moving walkway. They have measured the walkway to be $\ell = 42.5$ m long. The father has a stopwatch and times his daughter. First, the daughter walks with a constant speed, v_G, in the same direction as the conveyor. It takes $t_1 = 15.2$ s to reach the end of the walkway. Then, she turns around and walks with the same speed, v_G, relative to the conveyor as before, just this time in the opposite direction. The return leg takes $t_2 = 70.8$ s. What is the speed, v_W, of the walkway conveyor relative to the terminal, and with what speed, v_G was the girl walking?



displacement $\Delta x = x_f - x_i$ $= 0 - l$ $= -l$

1-d: $v_G + v_W = \frac{l}{t_1}$ –① ✓ $v_W - v_G = \frac{-l}{t_2}$ –②

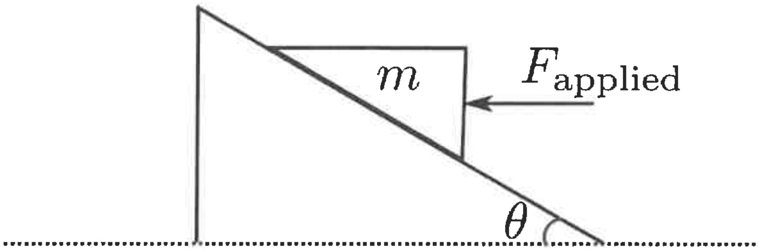
Two unknowns v_G and v_W. l, t_1 and t_2 are known. Solve between ① and ② ✓✓

$v_G = 1.70$ m/s ✓ and $v_W = 1.1$ m/s ✓

If you swapped the values, you can not receive full marks.

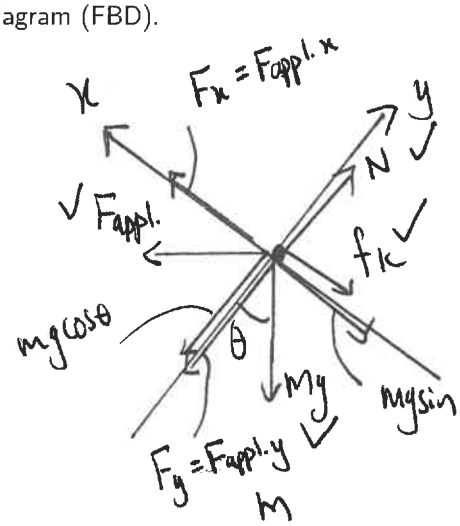
Question 8 [11]

A wedge of mass m is located on a plane that is inclined by an angle of θ with respect to the horizontal. A force of magnitude F_{applied} in the horizontal direction pushes on the wedge, as shown in the figure. The coefficient of kinetic friction between the wedge and the plane is μ_k and the coefficient of static friction between the wedge and the plane is μ_s. Find an expression for the acceleration of the wedge along the plane (your expression needs to only contain the given quantities).



Sketch your free-body diagram (FBD). (4) 4

There is no static friction as the block m is moving



Coordinate system must be shown

Components for Fappl. and mg!

(7) 7

y-components: $\sum F_y = 0$

$N - F_y - mg\cos\theta = 0$

$N = F_{appl}\sin\theta + mg\cos\theta$ ✓✓ $-(1)$

x-components: $\sum F_x = ma_x$ ✓

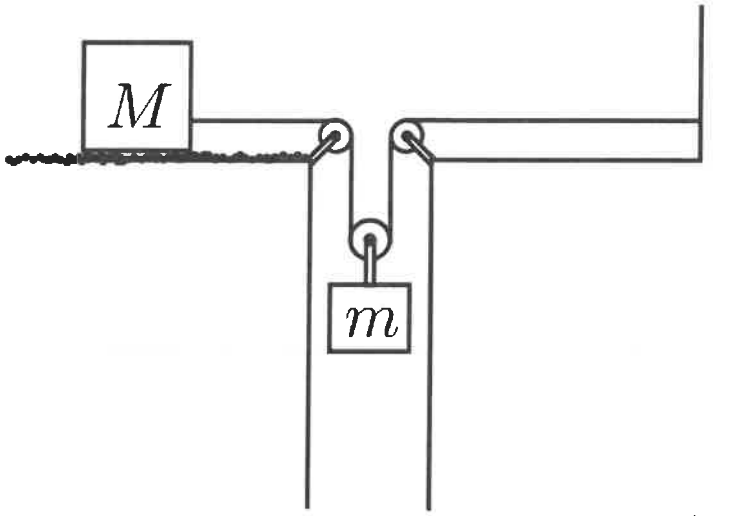
✓ $F_x - mg\sin\theta - f_k = ma_x$ $-(2)$

$F_{appl}\cos\theta$ ✓ $\quad$ $\mu_k N \rightarrow$ From (1)

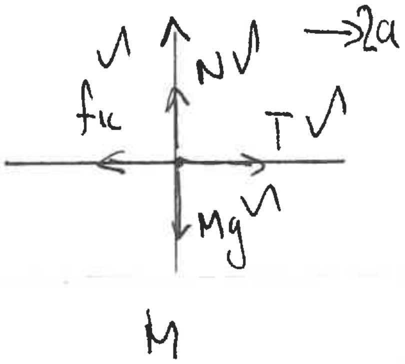
Find: $a_x = \frac{F_{appl}}{m}\left[\cos\theta - \mu_k \sin\theta\right] - g\left[\mu_k\cos\theta + \sin\theta\right]$ ✓✓

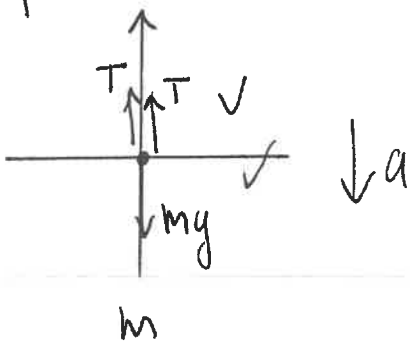
Question 9 [11]

In the figure below, when the mass of m is 3.0 kg, the acceleration of m is $a = 0.6$ m/s^2. If m is swapped out for another block of mass $m = 4.0$ kg, the acceleration of the block m is $a = 1.6$ m/s^2. Find the frictional force on M as well as its mass, if the surface it rests on is rough. You may ignore any friction on the pulleys as well its mass. The string is treated as massless. You need to solve the problem by making use of the component forces.



Sketch your free-body diagrams (FBDs). M has twice the acceleration of M (4) 4





For m: y-components: $\Sigma F_y = ma$ (7) 7

$2T - mg = ma$

$T = \frac{m}{2}(g-a)$ — ①

For block M: x-components Acceleration is 2a for block M, if the acceleration of block m is a ✓

$$\sum F_x = 2Ma$$

$$\Rightarrow f_k = T - 2Ma \quad -②$$

$a = a_1 = 0.6\,m/s^2$

Solve for T if $m = m_1 = 3\,kg$, $\Rightarrow T_1 = 13.8\,N$ ✓

" " T if $m = m_2 = 4\,kg$, $\Rightarrow T_2 = 16.4\,N$ ✓

$a = a_2 = 1.6\,m/s^2$

Substitute T_1 and T_2 into ② for the different values of a; solve for M and f_k. ✓✓

$M = 1.3\,kg$ ✓ and $f_k = 12.2\,N$ ✓

END OF TEST